

DIGITAL FORENSICS

EXAMINATION REPORT

National Gallery DC – Stamp Theft & Art Defacement Case

Subject Device: Tracy Sumtwelve's iPhone

Prepared by:

Ayham Belal Megdadi

Cybersecurity & Cloud Computing

Report Control	Value
Case ID	2012-07-15-National-Gallery
Evidence Image	tracy-phone-2012-07-15-final.E01
Evidence Type	Mobile device forensic image
Primary Subject	Tracy Sumtwelve
Examination Scope	Acquisition integrity, device profiling, email, SMS, images/EXIF, location data, timeline
Report Status	Final examination report – educational/case exercise
Classification	Confidential / Investigative Use

REPORT PURPOSE

This report documents the examination of a supplied forensic image and the reconstruction of digital evidence relevant to the National Gallery DC case. It is structured to reflect defensible digital-forensic reporting practices, while distinguishing observed artifacts from investigative interpretation.

Prepared and examined by: Ayham Belal Megdadi

Cybersecurity & Cloud Computing

Index

Document Control	4
Standards and Best-Practice Basis	4
1. Executive Summary	5
2. Case Background and Objectives	6
2.1 Examination Questions	6
2.2 Scope	6
2.3 Out of Scope	6
3. Evidence Identification and Integrity	7
3.1 Evidence Source	8
3.2 Integrity Controls	8
4. Examination Methodology	8
4.1 Tools and Techniques	8
4.2 Timestamp Handling	8
5. Device Profiling	9
5.1 Attribution Assessment	9
5.2 Email Account Inventory	9
6. Email and Attachment Analysis	11
6.1 Recruitment / Conspiracy Email	11
6.2 Insurance Documentation	11
6.3 Preparation Attachment	11
7. SMS Analysis	15
7.1 Financial Pressure	15
7.2 Coordination with Carry	15
7.3 Security Information Exchange	15
8. Image and EXIF Analysis	18
8.1 Photographic Reconnaissance	18
8.2 Geographic Corroboration	18
9. Location Analysis	19
9.1 Movement Correlation	19
10. Integrated Timeline	20
10.1 Timeline Interpretation	20
11. Evidence Correlation and Findings	21
11.1 Role Assessment	21
12. Conclusions	22
12.1 Evidence Quality	22
13. Recommendations	23
13.1 Preservation Priorities	23
14. Chain of Custody and Evidence Handling	24

14.1 Evidence Handling Statement	24
15. Evidence Register	24
15.1 Key Artifact Paths	24
16. Stamp Insurance Documentation	25
Appendix A – Standards Mapping.....	26
Reference Standards	26
Appendix B – Analyst Declaration.....	27

Document Control

Field	Entry
Document Title	Digital Forensics Examination Report – Tracy iPhone
Case	National Gallery DC – Stamp Theft & Art Defacement
Evidence Image	tracy-phone-2012-07-15-final.E01
Computed MD5	34c4888f095dc3241330462923f6fea5
Computed SHA-256	71aed05a86a753dec4ef4033ed7f52d6577ccb534ca0d1e83ffd27683e621607
Time Zone Context	America/New_York (Eastern), per examination notes
Primary Tool	Autopsy
Supporting Methods	SQLite analysis, EXIF metadata review, timeline correlation, coordinate verification
Evidence Integrity	Published MD5 and SHA-256 values match the computed values

Standards and Best-Practice Basis

This report is structured with reference to widely used digital-forensics guidance. NIST SP 800-86 provides a practical framework for forensic activities; ISO/IEC 27037 addresses identification, collection, acquisition and preservation of digital evidence; ISO/IEC 27042 addresses analysis and interpretation with emphasis on continuity, validity, reproducibility and repeatability; and SWGDE publishes discipline-specific best practices, including mobile-device evidence handling and forensic reporting.

Reference	How it is reflected in this report
NIST SP 800-86	Evidence identification, acquisition/integrity, examination, analysis, reporting and documentation.
ISO/IEC 27037:2012	Evidence integrity, preservation, acquisition context and traceability.
ISO/IEC 27042:2015	Separation of observations from interpretation, reproducibility, limitations and corroboration.
SWGDE best practices	Evidence handling, examination documentation, reporting structure and explicit limitations.

IMPORTANT

“Complies with” is not claimed as a formal certification. This report is aligned to the principles and reporting expectations described by the cited guidance, based on the evidence and process documentation supplied for this exercise.

1. Executive Summary

Examiner: **Ayham Belal Megdadi**

Role: **Digital Forensic Examiner**

Date: **30/05/2026**

The forensic examination and analysis performed by Ayham Belal Megdadi on the supplied forensic image of Tracy Sumtwelve's iPhone identified multiple independent artifact classes relevant to the National Gallery DC investigation. The strongest evidentiary pattern is created by the correlation of email artifacts, SMS communications, photographs with embedded GPS metadata, and iOS location databases.

EXECUTIVE FINDING

The examined evidence supports the assessment that the device was used in planning and coordinating activities connected to the National Gallery DC case and that the device was physically present at locations relevant to the investigation. The evidence is strongest where independent artifacts converge on the same event, time and location.

Finding	Primary Evidence	Assessment
Evidence integrity	MD5 + SHA-256 hash verification	High confidence
Device attribution	Apple support logs, account artifacts and device identifiers	High confidence
Planning / coordination	Email and SMS artifacts	High confidence
Target reconnaissance	Stamp photographs + EXIF metadata	High confidence
Physical presence at NGDC	Photo GPS + location artifacts	High confidence when considered jointly
Financial motive	School-cost SMS + insurance documentation	Supported; motive is interpretive
Insurance-fraud hypothesis	Insurance documents + financial pressure	Investigative hypothesis; not independently proven by the image

2. Case Background and Objectives

The supplied case material describes an investigation concerning alleged theft of valuable stamps from the National Gallery DC collection and coordinated art-defacement activity involving a flash-mob operation. **Tracy Sumtwelve** is identified as the primary subject and an NGDC insider.

The examination objective is to determine what relevant digital evidence can be recovered from the supplied iPhone image and to establish a defensible chronology of communications, reconnaissance, location and planning activity.

2.1 Examination Questions

- 1) Can the integrity of the forensic image be independently verified?
- 2) What device and user/account information can be established?
- 3) What relevant email communications and attachments are recoverable?
- 4) What SMS communications and contacts are relevant to the investigation?
- 5) Do photographs contain metadata that can establish time, location or subject matter?
- 6) Does the iOS location database corroborate the photographic and communications evidence?
- 7) Can the independent artifacts be correlated into a coherent investigative timeline?
- 8) What conclusions are supported, and what remains an inference or requires external corroboration?

2.2 Scope

- Forensic image integrity and source identification.
- iOS device profiling and account identification.
- Mail artifacts under **/vol5/mobile/Library/Mail/**.
- SMS database at **/vol5/mobile/Library/SMS/sms.db**.
- Camera media under **/vol5/mobile/Media/DCIM/100APPLE/**.
- Location artifacts under **/vol5/root/Library/Caches/locationd/consolidated.db**.
- Cross-artifact timeline reconstruction.

2.3 Out of Scope

- Independent examination of the original physical iPhone.
- Independent acquisition from external email providers or other suspects' devices.
- Independent authentication of insurance documents with the insurer.
- Independent verification of museum CCTV, access-control records or financial records.
- Legal determination of guilt or liability.

Examiner attribution: Ayham Belal Megdadi performed the examination, artifact analysis, cross-correlation, interpretation, and reporting presented in this document.

3. Evidence Identification and Integrity

The supplied forensic image is identified as tracy-phone-2012-07-15-final.E01. The examination notes state that the image was treated as intact and that the published cryptographic hashes matched the computed values before analysis.

Hash Algorithm	Computed	Published	Result
MD5	34c4888f095dc3241330462923f6fea5	34c4888f095dc3241330462923f6fea5	MATCH
SHA-256	71aed05a86a753dec4ef4033ed7f52d6577ccb534ca0d1e83ffd27683e621607	71aed05a86a753dec4ef4033ed7f52d6577ccb534ca0d1e83ffd27683e621607	MATCH

```

• MD5: 34c4888f095dc3241330462923f6fea5

C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>cd C:\Forensics\Tracy
The system cannot find the path specified.

C:\Users\DELL>cd c:\Users\DELL\Downloads

c:\Users\DELL\Downloads>certutil -hashfile tracy-phone-2012-07-15-final.E01 MD5
MD5 hash of tracy-phone-2012-07-15-final.E01:
34c4888f095dc3241330462923f6fea5
CertUtil: -hashfile command completed successfully.

```

```

• MD5: 34c4888f095dc3241330462923f6fea5
• SHA-256: 71aed05a86a753dec4ef4033ed7f52d6577ccb534ca0d1e83ffd27683e621607

C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>cd C:\Forensics\Tracy
The system cannot find the path specified.

C:\Users\DELL>cd c:\Users\DELL\Downloads

c:\Users\DELL\Downloads>certutil -hashfile tracy-phone-2012-07-15-final.E01 MD5
MD5 hash of tracy-phone-2012-07-15-final.E01:
34c4888f095dc3241330462923f6fea5
CertUtil: -hashfile command completed successfully.

c:\Users\DELL\Downloads>certutil -hashfile tracy-phone-2012-07-15-final.E01 SHA256
SHA256 hash of tracy-phone-2012-07-15-final.E01:
71aed05a86a753dec4ef4033ed7f52d6577ccb534ca0d1e83ffd27683e621607
CertUtil: -hashfile command completed successfully.

```

INTEGRITY STATUS: PASSED

The matching MD5 and SHA-256 values provide a documented integrity check for the supplied forensic image.

3.1 Evidence Source

Attribute	Value
Source type	Disk/forensic image of a mobile-device filesystem
Image filename	tracy-phone-2012-07-15-final.E01
Filesystem root used in analysis	/vol5/
Primary timezone	America/New_York (Eastern)
Integrity check	MD5 + SHA-256

3.2 Integrity Controls

1. Record the original evidence identifier and acquisition/image filename.
2. Calculate and record cryptographic hashes before examination.
3. Use a read-only or otherwise controlled forensic workflow.
4. Record relevant artifact paths, timestamps and identifiers.
5. Preserve the original image and create verified working copies as required.
6. Document any discrepancy or ambiguity rather than silently modifying source data.

4. Examination Methodology

The examination followed an artifact-first and correlation-based workflow. The supplied case material records use of Autopsy for filesystem and artifact review, SQLite analysis for iOS databases, coordinate verification, and timeline reconstruction.

Phase	Activity	Primary Output
1	Evidence integrity verification	Hash match / integrity status
2	Filesystem examination	Directory and artifact inventory
3	Device profiling	Device/account identifiers
4	Email examination	Messages, accounts and attachments
5	SMS examination	Messages, contacts and timestamps
6	Image/EXIF examination	Photo metadata and GPS
7	Location analysis	WiFi/cell location records
8	Correlation	Cross-artifact relationships
9	Timeline reconstruction	Master chronology
10	Reporting	Findings, limitations and recommendations

4.1 Tools and Techniques

Tool / Technique	Purpose
Autopsy	Filesystem navigation, artifact review, email/SMS examination, EXIF and evidence organization
SQLite analysis	Examination of sms.db and consolidated.db
EXIF metadata review	Camera timestamps, device metadata and GPS coordinates
Coordinate verification	Independent geographic interpretation of extracted coordinates
Timeline correlation	Cross-source ordering and event reconstruction

4.2 Timestamp Handling

For SMS artifacts, the supplied case material documents the iOS epoch conversion using 1 January 2001 as the reference point. The documented conversion is: Unix timestamp = iOS timestamp + 978,307,200, followed by local-time interpretation.

5. Device Profiling

The device profile was derived from Apple support/log artifacts and related iOS records. The source material identifies the device as an iPhone 1,2 running iPhone OS 4.2.1 (8C148).

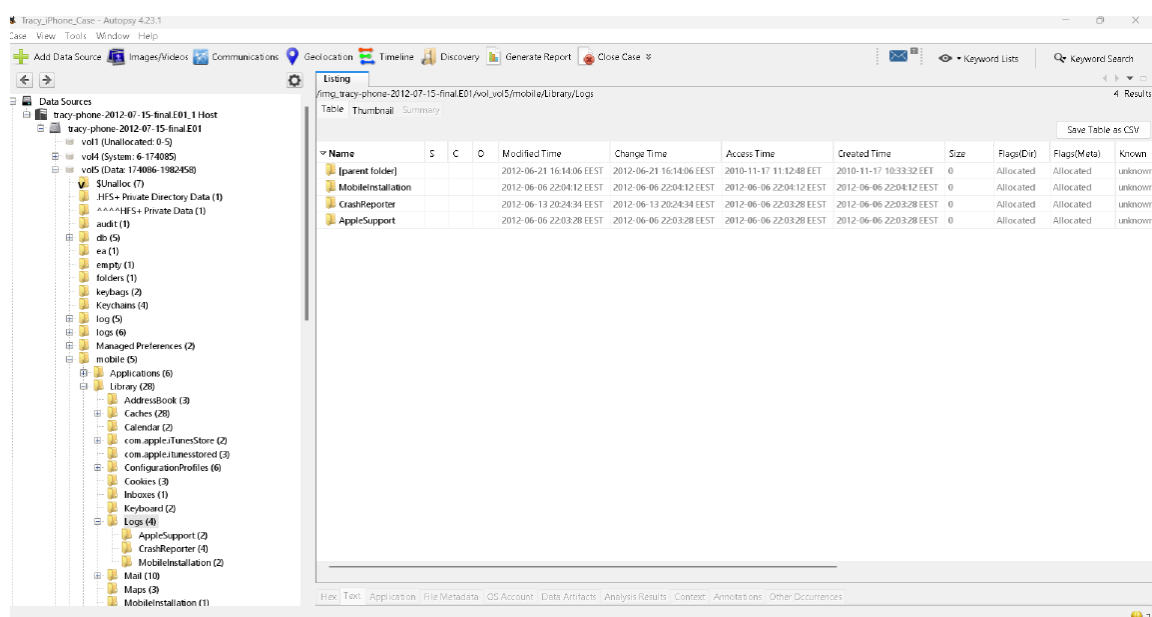
Field	Examination Finding	Source Path / Context
Device Model	iPhone 1,2	/vol5/mobile/Library/Logs/AppleSupport/general.log
OS	iPhone OS 4.2.1 (8C148)	AppleSupport/general.log
Install / creation date	June 6, 2012 – 12:03:28 UTC-0700	AppleSupport/general.log
Owner name	Tracy Sumtwelve	AppleSupport/general.log
Phone number	1(703)340-9661	Device/log artifacts
IMEI	010221003735398	Activation/lockdown artifacts
ICCID	89014103255195342366	Lockdown artifacts
Primary email	tracysumtwelve@gmail.com	Mail artifacts
Work email	tracy.sumtwelve@nationalgallerydc.org	Mail artifacts
Alias email	coralbluetwo@hotmail.com	POP3 mailbox artifacts

5.1 Attribution Assessment

The combination of device-level identifiers, owner name, phone number, personal/work accounts and mailbox artifacts provides strong attribution of the examined filesystem to the subject described in the case materials. Attribution is nevertheless an evidentiary assessment, not an absolute statement that every action represented in an account was personally performed by the named individual.

5.2 Email Account Inventory

Account	Protocol	Role / Relevance
tracy.sumtwelve@nationalgallerydc.org	IMAP	Work account associated with NGDC
tracy.sumtwelve@gmail.com	IMAP	Personal account
coralbluetwo@hotmail.com	POP3	Alias account containing investigation-relevant material



File Explorer showing the file system structure of a mobile device. The left pane shows the hierarchy: **Skinallot (7)**, **JFS+ Private Directory Data (1)**, ******JFS+ Private Data (1)**, **audit (1)**, **db (5)**, **ea (1)**, **empty (1)**, **folders (1)**, **keybags (2)**, **keychains (4)**, **log (5)**, **logs (6)**, **Managed Preferences (2)**, **mobile (5)**, **Applications (6)**, **Library (28)**, **AddressBook (3)**, **Caches (28)**, **Calendar (2)**, **com.apple.iTunesStore (2)**, **com.apple.iTunesStore (3)**, **ConfigurationProfiles (6)**, **Cookies (3)**, **Inboxes (1)**, **Keyboard (2)**, **Logs (4)**, **AppleSupport (2)**, **CrashReporter (4)**, **MobileInstallation (2)**, **Mail (10)**, **Maps (3)**, **MobileInstallation (1)**, **Notes (3)**, **Preferences (42)**, **Safari (6)**, **SMS (3)**, **Spotlight (2)**, **SpringBoard (7)**.

The right pane shows the **Listing** of files in the **AppleSupport** directory. The table lists files with columns: **Name**, **S**, **C**, **D**, **Modified Time**, **Change Time**, **Access Time**, **Created Time**, **Size**, **Flags(Dir)**, **Flags(Meta)**, and **Known**.

Name	S	C	D	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known
general.log		0		2012-07-10 23:50:27 EEST	2012-07-10 23:50:27 EEST	2012-06-06 22:03:28 EEST	2012-06-06 22:03:28 EEST	2102	Allocated	Allocated	unknown
(parent folder)				2012-06-06 22:04:12 EEST	2012-06-06 22:04:12 EEST	2012-06-06 22:03:28 EEST	2012-06-06 22:03:28 EEST	0	Allocated	Allocated	unknown

The right pane shows the **File Metadata** of the **general.log** file. The metadata includes:

- Device Software Diagnostic Log
- Version: 3
- OS-Version: iPhone OS 4.2.1 (8C148)
- Model: iPhone1,2
- Serial Number: B50044827H
- Created: 6/6/2012 12:03:28 -0700

The **Strings** section shows a list of strings extracted from the file, including various system logs and timestamps.

The right pane shows the **Listing** of files in the **AppleSupport** directory. The table lists files with columns: **Name**, **S**, **C**, **D**, **Modified Time**, **Change Time**, **Access Time**, **Created Time**, **Size**, **Flags(Dir)**, **Flags(Meta)**, and **Known**.

Name	S	C	D	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known
general.log		0		2012-07-10 23:50:27 EEST	2012-07-10 23:50:27 EEST	2012-06-06 22:03:28 EEST	2012-06-06 22:03:28 EEST	2102	Allocated	Allocated	unknown
(parent folder)				2012-06-06 22:04:12 EEST	2012-06-06 22:04:12 EEST	2012-06-06 22:03:28 EEST	2012-06-06 22:03:28 EEST	0	Allocated	Allocated	unknown

6. Email and Attachment Analysis

Email artifacts were examined under **/vol5/mobile/Library/Mail/**. The alias mailbox **coralbluetwo@hotmail.com** contained investigation-relevant material, including planning correspondence and attachments. The supplied case report records mailbox statistics and specific **.emlx** evidence locations.

Artifact Category	Finding	Forensic Significance
Alias mailbox	POP3 coralbluetwo@hotmail.com	Separates sensitive communications from the subject's named accounts
Planning correspondence	Messages involving Pat and King	Supports organized planning
Insurance documents	Three PDF documents packaged as documents.zip	Shows knowledge of insured stamp lots and values
needs.txt	Operational preparation attachment	Supports advance preparation
Email evidence paths	/vol5/mobile/Library/Mail/POP-coralbluetwo.../INBOX	Provides reproducible artifact location

6.1 Recruitment / Conspiracy Email

The supplied case material identifies an email dated July 6, 2012 at 11:49:31 -0400 from **patsumtwelve@gmail.com** to **throne1966@hotmail.com**, with coralbluetwo@hotmail.com copied. The email explicitly describes a planned heist at the National Gallery and attempts to recruit **"King."**

CORROBORATING FACT

Tracy's alias account is recorded as a CC recipient on the recruitment email. This is stronger than an inference based solely on the existence of the alias mailbox because it directly associates the account with the planning communication.

6.2 Insurance Documentation

Three insurance documents were recovered as attachments. The supplied case report itemizes nine lots and records a total estimated value of \$260,000. Your examination notes separately report a total of \$270,000. Because the source materials disagree, this report preserves both values and treats the discrepancy as a data-quality issue requiring source-document verification.

Source	Reported Total	Treatment in this Report
CASE TRACY.pdf	\$260,000	Primary case-report value
Examination notes supplied with the request	\$270,000	Recorded as an alternate reported value

6.3 Preparation Attachment

The attachment named **needs.txt** contains a list of equipment and preparation items associated with the planned operation. In this report, the list is treated as evidence of preparation and intent; individual items are not interpreted as proof that any particular act was actually carried out.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
metadata.plist			0	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST
(parent folder)				2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	2010-11-17 11:12:48 EET	2010-11-17 10:33:32 EET
Vault				2012-06-06 22:04:40 EEST	2012-06-06 22:04:40 EEST	2012-06-06 22:04:40 EEST	2012-06-06 22:04:40 EEST
Protected Index			0	2012-07-12 23:40:26 EEST	2012-07-12 23:40:26 EEST	2012-06-06 22:04:40 EEST	2012-06-06 22:04:40 EEST
POP-coralbluetwo@hotmail.com@pop3.live.com				2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:50:34 EEST	2012-07-12 21:50:34 EEST
Mailboxes				2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST
IMAP-tracysumtwelve@gmail.com@imap.gmail.com				2012-06-20 16:58:56 EEST	2012-06-20 16:58:56 EEST	2012-06-13 21:39:50 EEST	2012-06-13 21:39:50 EEST
IMAP-tracysumtwelve@nationalgallerydc.org@pod51018.outlook.com				2012-07-12 21:47:05 EEST	2012-07-12 21:47:05 EEST	2012-07-12 21:47:05 EEST	2012-07-12 21:47:05 EEST
Envelope Index			0	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	2012-06-06 22:04:05 EEST	2012-06-06 22:04:05 EEST
AutoFetchEnabled			0	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST

Listing
/img_iphone-2012-07-15-final/E01/vol5/mobile/Library/Mail

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags/D
metastapilot			0	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	539	Allocates
parent folder				2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	0	Allocates
Vault				2012-06-06 22:04:40 EEST	2012-06-06 22:04:40 EEST	2012-06-06 22:04:40 EEST	2012-06-06 22:04:40 EEST	311296	Allocates
Protected Index			0	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	0	Allocates
POP-corabluetwo@hotmail.com@pop3.live.com				2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	0	Allocates
Mailboxes				2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	0	Allocates
IMAP-tracysumtwelve@gmail.com@imap.gmail.com				2012-06-20 16:58:56 EEST	2012-06-20 16:58:56 EEST	2012-06-20 16:58:56 EEST	2012-06-20 16:58:56 EEST	0	Allocates
IMAP-tracysumtwelve@hotmail.com@pop3				2012-07-12 21:47:05 EEST	2012-07-12 21:47:05 EEST	2012-07-12 21:47:05 EEST	2012-07-12 21:47:05 EEST	0	Allocates
Envelope Index			0	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	126976	Allocates
AutofetchEnabled			0	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	42	Allocates

Listing
/img_iphone-2012-07-15-final/E01/vol5/mobile/Library/Mail

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags/D
metastapilot			0	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	539	Allocates
parent folder				2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	0	Allocates
Vault				2012-06-06 22:04:40 EEST	2012-06-06 22:04:40 EEST	2012-06-06 22:04:40 EEST	2012-06-06 22:04:40 EEST	311296	Allocates
Protected Index			0	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	0	Allocates
POP-corabluetwo@hotmail.com@pop3.live.com				2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	0	Allocates
Mailboxes				2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	2012-06-13 21:50:12 EEST	0	Allocates
IMAP-tracysumtwelve@gmail.com@imap.gmail.com				2012-06-20 16:58:56 EEST	2012-06-20 16:58:56 EEST	2012-06-20 16:58:56 EEST	2012-06-20 16:58:56 EEST	0	Allocates
IMAP-tracysumtwelve@hotmail.com@pop3				2012-07-12 21:47:05 EEST	2012-07-12 21:47:05 EEST	2012-07-12 21:47:05 EEST	2012-07-12 21:47:05 EEST	0	Allocates
Envelope Index			0	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	126976	Allocates
AutofetchEnabled			0	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	2012-07-15 12:55:06 EEST	42	Allocates

Listing
/img_iphone-2012-07-15-final/E01/vol5/mobile/Library/Mail/POP-corabluetwo@hotmail.com@pop3.live.com/Deleted Messages/Inbox

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags/D
parent folder				2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	0	Allocates
Messages				2012-07-12 21:51:11 EEST	2012-07-12 21:51:11 EEST	2012-07-12 21:51:11 EEST	2012-07-12 21:51:11 EEST	0	Allocates

Listing
/img_iphone-2012-07-15-final/E01/vol5/mobile/Library/Mail/POP-corabluetwo@hotmail.com@pop3.live.com/Deleted Messages/Inbox/Deleted Messages

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags/D
parent folder				2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	2012-07-12 21:56:45 EEST	0	Allocates
Deleted Messages				2012-07-12 21:51:11 EEST	2012-07-12 21:51:11 EEST	2012-07-12 21:51:11 EEST	2012-07-12 21:51:11 EEST	0	Allocates

Listing

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size
[parent folder]				2012-06-20 16:58:56 EEST	2012-06-20 16:58:56 EEST	2012-06-20 16:58:56 EEST	2012-06-20 16:58:56 EEST	0
14872467-18E-474D-A1E6-A78EE1EEAE2.emb	0			2012-07-05 21:21:17 EEST	2012-07-05 21:21:17 EEST	2012-07-05 21:21:17 EEST	2012-07-05 21:21:17 EEST	11
B107974-1A38-414A-A944-5414D1557892.emb	0			2012-07-12 21:32:25 EEST	2012-07-12 21:32:25 EEST	2012-07-12 21:32:25 EEST	2012-07-12 21:32:25 EEST	11
965C2E29-C540-4B57-928A-D9E704DC0F832.emb	0			2012-07-10 18:59:44 EEST	2012-07-10 18:59:44 EEST	2012-07-10 18:59:44 EEST	2012-07-10 18:59:44 EEST	11
7A457E71-2E23-442E-87A0-D9029651M492.emb	0			2012-07-12 21:32:22 EEST	2012-07-12 21:32:22 EEST	2012-07-12 21:32:22 EEST	2012-07-12 21:32:22 EEST	11
6314C1A3-A26E-45E9-9222-C72F9A78D0A82.emb	0			2012-07-05 21:21:19 EEST	2012-07-05 21:21:19 EEST	2012-07-05 21:21:19 EEST	2012-07-05 21:21:19 EEST	11
34EC8E8-25F3-4136-804E-EF61CEA5652.emb	0			2012-07-10 18:59:48 EEST	2012-07-10 18:59:48 EEST	2012-07-10 18:59:48 EEST	2012-07-10 18:59:48 EEST	11

Listing

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)
DS_Store				2012-07-09 09:42:58 EEST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	6148	Allocated	Allocated
Stamp insurance 1.pdf	0			2012-07-06 09:39:52 EEST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	189553	Allocated	Allocated
Stamp insurance 2.pdf	0			2012-07-06 09:39:52 EEST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	183592	Allocated	Allocated
Stamp insurance 3.pdf	0			2012-07-06 09:39:52 EEST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	182641	Allocated	Allocated

NATIONAL GALLERY OF ART
WASHINGTON

Memorandum of Insurance Assurance:

TO: MyStamp Collections

In Regards to items owned or on loan to the National Gallery of Art, Washington from MyStamp Collections from the period of May 12, 2012 to May 12, 2013, all items in aforementioned list will be in the custody of National Gallery of Art, and therefore fiscal responsibility of said items is covered by the same. As per agreement between both parties the following items are covered by Avesome Insurance Inc. Policy #23854 for the following amounts under any circumstances leading to the degradation of said items excepting normal wear of display.

Lot # 1. Douglas MacArthur	\$35,000.00
Lot # 2. Nederland	\$30,000.00
Lot # 3. Mongolia	\$24,000.00

Terms do not cover period of transport from or to the National Gallery.

D'Mann
President National Gallery DC



NATIONAL GALLERY DC
WASHINGTON



Memorandum of Insurance Assurance:

TO: MyiStamp Collections

In Regards to items owned or on loan to the National Gallery DC, Washington from MyiStamp Collections from the period of May 12, 2012 to May 12, 2013, all items in aforementioned list will be in the custody of National Gallery DC and therefore fiscal responsibility of said items is carried by the same. As per agreement between both parties the following items are covered by Awesomelnsurance Inc. Policy X23654 for the following amounts under any circumstances leading to the degradation of said items excepting normal wear of display.

Lot # 11. Woman's Profile	\$51,000.00
Lot # 12. Stamp of Kazakhstan	\$29,000.00
Lot# 13. 1929 Nepal	\$27,000.00

Terms do not cover period of transport from or to the National Gallery.

D'Mann

President National Gallery DC

The screenshot shows a forensic tool interface. On the left is a file system tree with folders like 'Calendar (2)', 'com.apple.iTunesStore (2)', 'ConfigurationProfiles (6)', 'Cookies (3)', 'Downloads (1)', 'Keyboard (2)', 'Logs (4)', 'AppleSupport (2)', 'CrashReporter (4)', 'MobileInstallation (2)', 'Mail (10)', 'IMAP-tracysumtweive@nationalgallerydc.org@pod51018.ox', 'IMAP-tracysumtweive@gmail.com@imap.gmail.com (3)', 'iHBOXimapbox (2)', 'Messages (7)', 'Mailboxes (2)', 'POP-coralbluetwo@hotmail.com@pop3.live.com (4)', 'Deleted Messages.mbox (2)', 'Messages (2)', 'iHBOXinbox (3)', 'Attachments (4)', '60 (2)', '2 (2)', '61 (3)', '2 (2)', 'documents.zip (1)', 'docs (4)', '3 (2)', 'docs.zip (2)', 'iHBOX (1)', 'docs (2)', 'docs (4)', 'Stamp insurance 1.pdf (1)', 'Stamp insurance 2.pdf (1)', 'Stamp insurance 3.pdf (1)', and '62 (2)'. On the right is a table with the following data:

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Rights(M)
needs.txt		0		2012-07-12 21:51:14 EEST	2012-07-12 21:51:14 EEST	2012-07-12 21:51:14 EEST	2012-07-12 21:51:14 EEST	81992	Allocated	Allocate
[parent folder]				2012-07-12 21:51:14 EEST	2012-07-12 21:51:14 EEST	2012-07-12 21:51:14 EEST	2012-07-12 21:51:14 EEST	0	Allocated	Allocate

7. SMS Analysis

The primary SMS database was identified as **/vol5/mobile/Library/SMS/sms.db**. The supplied examination notes report 25 message entries, while the supplied case report's evidence-location section reports 24 relevant messages. This discrepancy is documented in Section 13.

Contact	Identifier	Role in Case Material	Key Evidence
Pat	+15713083236	Brother / planning contact	Family and planning communications
Terry	+17038296071	Daughter	Financial-pressure discussion
Carry	+12027252124	Flash-mob organizer	Meeting and coordination messages
Unknown / Spam	+12069100932	Unresolved	No material finding established in supplied notes

7.1 Financial Pressure

A July 3, 2012 SMS exchange between Tracy and Terry discusses difficulty affording school tuition. The supplied case report interprets this as evidence of financial pressure. In this report, the exchange is treated as support for a possible financial motive, but not as proof of a criminal act.

7.2 Coordination with Carry

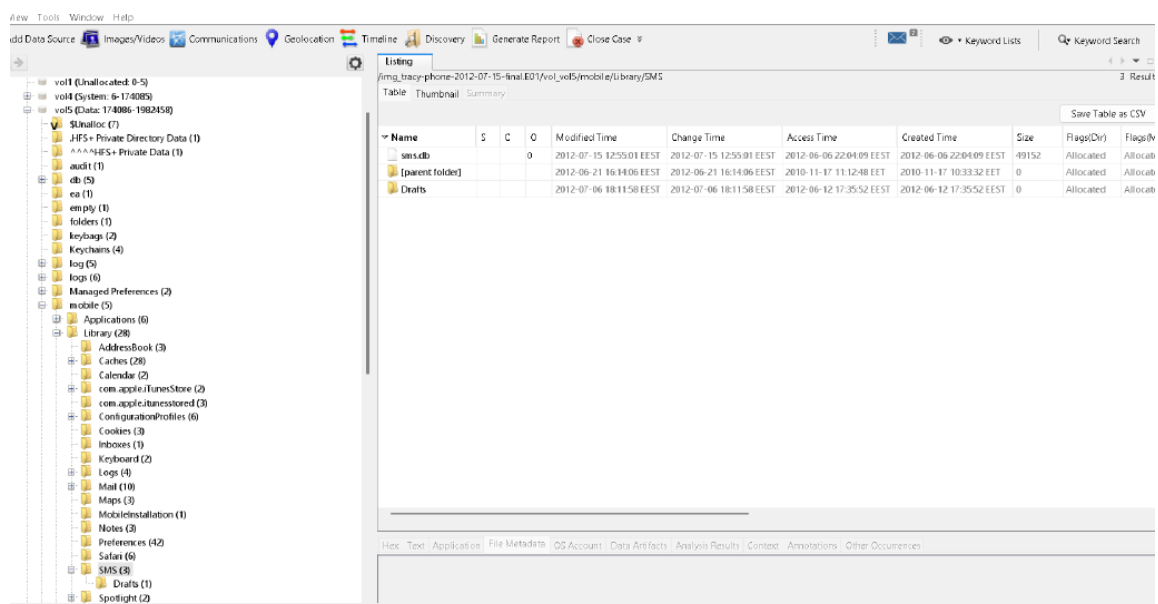
SMS messages around July 5–6 document arrangements to meet at **Bubba's Grill**, including a message from **Tracy** stating that she had a table inside. Later messages on July 11 document an equipment transfer involving a tablet and indicate that Tracy would take the device **into NGDC**.

7.3 Security Information Exchange

The case material records later communication concerning the flash-mob operation and security shift information. The significance of these messages is strengthened by their timing and their correlation with the subject's work at NGDC.

CORRELATION

SMS evidence is most probative when correlated with location records and email artifacts. The report therefore avoids treating the wording of individual messages as standalone proof of the entire alleged operation.



View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

/img_tracy-phone-2012-07-15-final.E01/vol_vols/mobile/Library/SMS

3 Results

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(M)
sms.db			0	2012-07-15 12:55:01 EEST	2012-07-15 12:55:01 EEST	2012-06-06 22:04:09 EEST	2012-06-06 22:04:09 EEST	49152	Allocated	Allocated
[parent folder]				2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	2010-11-17 11:12:48 EET	2010-11-17 10:33:32 EET	0	Allocated	Allocated
Drafts				2012-07-06 18:11:58 EEST	2012-07-06 18:11:58 EEST	2012-06-12 17:35:52 EEST	2012-06-12 17:35:52 EEST	0	Allocated	Allocated

Save Table as CSV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Table message 25 entries Page 1 of 1 Export to CSV

ROWID	address	date	text	flags	replace	svc_center	group_id
6	+15713083236	1339536304	What are you up to this weekend?	2	0		3
7	+17038296071	1339608628	I'm going out with dad after school for pizza! Thought I'd let you know if you planned to cook	2	0		4
8	+15713083236	1339612238	I don't have any big plans. How about you?	3	0		3
9	+17038296071	1339612426	Ok, sounds good.	3	0		4
12	+17038296071	1341322911	Hey honey, I'm not sure if we can afford ProRock anymore... What do you think about maybe	3	0		4
13	+17038296071	1341324272	moving schools at this point would be the worst! I would rather live with dad and stay at	2	0		4
14	+12027252124	1341512303	Sounds good! Let's shoot for one at Bubba's grill	2	0		5
15	+12027252124	1341512426	Okay that sounds great. See you there	3	0		5
16	+15713083236	1341580939	Hey can you give me a call	3	0		3
17	+15713083236	1341587317	Sis I'm really busy can we do this later	2	0		3
18	+15713083236	1341587514	No pat this is important I need you to call me soon	3	0		3
19	+15713083236	1341587611	Ok ok I'll call in 5	2	0		3
20	+12027252124	1341592036	I have a table inside	2	0		5
21	+12027252124	1341592070	Okay brt	3	0		5
22	+12069100932	1341689795	Congratulations, your entry in last month's drawing won you a FREE \$1,000 Target Giftcard! Ent 2	0			6

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

/img_tracy-phone-2012-07-15-final.E01/vol_vols/mobile/Library/SMS

3 Results

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(M)
sms.db			0	2012-07-15 12:55:01 EEST	2012-07-15 12:55:01 EEST	2012-06-06 22:04:09 EEST	2012-06-06 22:04:09 EEST	49152	Allocated	Allocated
[parent folder]				2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	2010-11-17 11:12:48 EET	2010-11-17 10:33:32 EET	0	Allocated	Allocated
Drafts				2012-07-06 18:11:58 EEST	2012-07-06 18:11:58 EEST	2012-06-12 17:35:52 EEST	2012-06-12 17:35:52 EEST	0	Allocated	Allocated

Save Table as CSV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Table group_member 4 entries Page 1 of 1 Export to CSV

ROWID	group_id	address	country
3	3	+15713083236	us
4	4	+17038296071	us
5	5	+12027252124	us
6	6	+12069100932	us

View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

/img_tracy-phone-2012-07-15-final.E01/vol_vols/mobile/Library/SMS

3 Results

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(M)
sms.db			0	2012-07-15 12:55:01 EEST	2012-07-15 12:55:01 EEST	2012-06-06 22:04:09 EEST	2012-06-06 22:04:09 EEST	49152	Allocated	Allocated
[parent folder]				2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	2010-11-17 11:12:48 EET	2010-11-17 10:33:32 EET	0	Allocated	Allocated
Drafts				2012-07-06 18:11:58 EEST	2012-07-06 18:11:58 EEST	2012-06-12 17:35:52 EEST	2012-06-12 17:35:52 EEST	0	Allocated	Allocated

Save Table as CSV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Table _SQLiteDatabaseY... 8 entries Page 1 of 1 Export to CSV

key	value
counter_last...	0
_ClientVersi...	11
_Uniqueliden...	4340118E-B575-4461-9099-8B5EE8SECAC
counter_out...	12
counter_out...	12
_CPRecord...	123
counter_in_a...	18
counter_in_b...	18

dd Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case %

Listing
/img_tracy-phone-2012-07-15-final.E01/vol5/mobile/Library/SMS 3 Results

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags/Dir
sms.db			0	2012-07-15 12:55:01 EEST	2012-07-15 12:55:01 EEST	2012-06-06 22:04:09 EEST	2012-06-06 22:04:09 EEST	49152	Allocated
[parent folder]				2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	2010-11-17 11:12:48 EET	2010-11-17 10:33:32 EET	0	Allocated
Drafts				2012-07-06 18:11:58 EEST	2012-07-06 18:11:58 EEST	2012-06-12 17:35:52 EEST	2012-06-12 17:35:52 EEST	0	Allocated

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Table _SQLiteDatabasePR... 8 entries Page 1 of 1 Export to CSV

key	value
counter_last_reset	0
ClientVersion	11
_UniqueIdentifier	4340110E-B575-4461-9099-48B5E185ECAC
counter_out_all	12
counter_out_lifetime	12
_CPRecordSequenceNumber	123
counter_in_all	18
counter_in_lifetime	18

Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case %

Listing
/img_tracy-phone-2012-07-15-final.E01/vol5/mobile/Library/SMS 3 Results

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags/Dir
Drafts				2012-07-06 18:11:58 EEST	2012-07-06 18:11:58 EEST	2012-06-12 17:35:52 EEST	2012-06-12 17:35:52 EEST	0	Allocated
[parent folder]				2012-06-21 16:14:06 EEST	2012-06-21 16:14:06 EEST	2010-11-17 11:12:48 EET	2010-11-17 10:33:32 EET	0	Allocated
sms.db	0			2012-07-15 12:55:01 EEST	2012-07-15 12:55:01 EEST	2012-06-06 22:04:09 EEST	2012-06-06 22:04:09 EEST	49152	Allocated

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Table message 25 entries Page 1 of 1 Export to CSV

ROWID	address	date	text	flags	replace	svc_center
6	+15713083236	1339536304	What are you up to this weekend?	2	0	
7	+17038296071	1339608628	I'm going out with dad after school for pizzal Thought I'd let you know if you planned to cook	2	0	
8	+15713083236	1339612238	I don't have any big plans. How about you?	3	0	
9	+17038296071	1339612426	Ok, sounds good.	3	0	
12	+17038296071	1341322911	Hey honey, I'm not sure if we can afford Prutrock anymore... What do you think about maybe s	3	0	
13	+17038296071	1341324272	moving schools at this point would be the worst I would rather live with dad and stay at prut	2	0	
14	+12027252124	1341512303	Sounds good let's shoot for one at Bubba's grill	2	0	
15	+12027252124	1341512426	Okay that sounds great. See you there	3	0	
16	+15713083236	1341506939	Hey can you give me a call	3	0	
17	+15713083236	1341507317	Sis I'm really busy can we can do this later	2	0	
18	+15713083236	1341507514	No pat this is important I need you to call me soon	3	0	
19	+15713083236	1341507611	Ok ok I'll call in 5	2	0	
20	+12027252124	1341592036	I have a table inside	2	0	
21	+12027252124	1341592070	Okay birt	3	0	
22	+12069100932	1341689795	Congratulations, your entry in last months drawing won you a FREE \$1,000 Target Giftcard! Ent	2	0	

8. Image and EXIF Analysis

The camera media was examined under **/vol5/mobile/Media/DCIM/100APPLE/**. The supplied examination notes identify **IMG_0050.JPG** as a primary forensic image and report embedded camera and GPS metadata.

EXIF Field	Reported Value
Filename	IMG_0050.JPG
Camera make	Apple
Camera model	iPhone 3G
Capture time	2012-07-08 12:41:53
Latitude	38.89166666666666° N
Longitude	-77.0235° W
Altitude	78.04526748971193 m
Content	Photograph of a postage stamp; Nepalese script reported visible

8.1 Photographic Reconnaissance

The case material records a series of stamp photographs captured on July 8, 2012, including a 1929 Nepal stamp, Stamp of Kazakhstan, Douglas MacArthur stamps, Armed Forces Reserve and BradyCo. material. The photographs are described as having GPS metadata placing the device within the National Gallery DC area.

8.2 Geographic Corroboration

The extracted coordinates **38.8917° N, -77.0235° W** were interpreted in the supplied analysis as being inside the **National Gallery of Art / NGDC complex**. This location is independently corroborated by iOS WiFi/location artifacts in the supplied case findings.

EVIDENCE STRENGTH

The combination of image timestamp + embedded GPS + subject matter + independent location-database evidence is materially stronger than any one of those artifacts considered alone.

The screenshot displays a forensic analysis interface. The top pane shows a file list with columns for Name, S, C, D, Modified Time, Change Time, Access Time, Created Time, Size, and Flags. The bottom pane shows a detailed view of a postage stamp image, which is a 1929 Nepal stamp featuring a central figure and text in Nepalese script. The interface includes various toolbars and navigation options.

9. Location Analysis

The location database was identified as **/vol5/root/Library/Caches/locationd/consolidated.db**. The supplied case material reports more than 2,005 WiFi-based location records and 376 cell-tower records, across tables including WifiLocation, CellLocation and LocationHarvest.

Date / Time	Coordinates	Reported Location	Relevance
2012-06-13 19:01:22	38.88055896, -77.11553561	900 N Glebe Rd, Arlington, VA	Pre-incident movement
2012-07-02 16:19:24	38.87990736, -77.11460858	800 N Glebe Rd, Arlington, VA	Movement pattern
2012-07-05 16:32:47	38.88054466, -77.11439651	801 N Glebe Rd, Arlington, VA	Correlates with planning period
2012-07-10 16:46:29	38.82760035, -77.08524417	1715 Centre Plaza, Alexandria, VA	Later movement
2012-07-08 ~12:35-12:41	38.8917, -77.0235	NGDC gallery area	Corroborates photo GPS

9.1 Movement Correlation

- Location records place the device in the DC/Arlington area during the planning period.
- The July 8 location evidence is consistent with the EXIF GPS coordinates associated with stamp photographs.
- The location evidence provides corroboration rather than a standalone identification of the person holding the device.
- Location accuracy and collection methodology should be preserved when presenting these findings externally.

The screenshot displays a forensic analysis interface. On the left, a file tree shows the directory structure of a device, including folders like 'YouTube', 'Media', 'Books', 'DCIM', 'JMS', 'Downloads', 'iTunes_Control', 'Photos', 'Podcasts', 'Purchases', 'Recordings', 'Safari', 'MobileDevice', 'preferences', 'root', 'Library', 'Backups', 'locationd', 'ephemera', 'Lockdown', 'run', 'spool', 'tmp', 'var', 'wireless', and 'vcl6'. The right pane shows a table of location data. The table has columns for Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, Flags(Dir), Flags(Meta), and K. The table lists files like 'parent folder', 'cache.plist', 'clients.plist', and 'consolidated.db'. Below this, a table titled 'Table: WifiLocation' shows 2005 entries with columns for MAC, Timestamp, Latitude, Longitude, Horizontal, Altitude, Vertical, Speed, Course, and Confidence.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	K
parent folder				2012-06-12 22:28:28 EEST	2012-06-12 22:28:28 EEST	2012-06-06 22:03:30 EEST	2012-06-06 22:03:30 EEST	0	Allocated	Allocated	un
cache.plist			0	2012-07-15 12:55:00 EEST	2012-07-15 12:55:00 EEST	2012-06-06 22:04:14 EEST	2012-06-06 22:04:14 EEST	883	Allocated	Allocated	un
clients.plist			0	2012-07-10 20:08:00 EEST	2012-07-10 20:08:00 EEST	2012-06-12 22:59:39 EEST	2012-06-12 22:59:39 EEST	718	Allocated	Allocated	un
consolidated.db			0	2012-07-12 21:30:47 EEST	2012-07-12 21:30:47 EEST	2012-06-06 22:04:13 EEST	2012-06-06 22:04:13 EEST	434176	Allocated	Allocated	un

MAC	Timestamp	Latitude	Longitude	Horizontal	Altitude	Vertical	Speed	Course	Confiden..
441ea1f4d8	3.6130688247 37 E5E8	38.88055896	-77.11553561	281.0	96.0	19.0	-1.0	-1.0	50
0235e6b6d4	3.6130688247 37 E5E8	38.88106283	-77.11533038	68.0	113.0	13.0	-1.0	-1.0	50
026dbac1c	3.6130688247 37 E5E8	38.88055346	-77.11595332	42.0	103.0	23.0	-1.0	-1.0	50
c0c1c0f156	3.6130688247 37 E5E8	38.88093715	-77.11640596	42.0	134.0	9.0	-1.0	-1.0	50
e0469a3f1	3.6130688247 37 E5E8	38.87996816	-77.11601394	42.0	104.0	38.0	-1.0	-1.0	50
5475db0a5c	3.6130688247 37 E5E8	38.88138395	-77.11556851	48.0	133.0	43.0	-1.0	-1.0	50
026dbac19	3.6130688247 37 E5E8	38.88139647	-77.11564362	42.0	111.0	31.0	-1.0	-1.0	50
026dbac19	3.6130688247 37 E5E8	38.87974703	-77.11598318	42.0	101.0	21.0	-1.0	-1.0	50
026dbac19	3.6130688247 37 E5E8	38.87969022	-77.11548959	42.0	99.0	17.0	-1.0	-1.0	50
026f3f8b08	3.6130688247 37 E5E8	38.87970983	-77.11530274	42.0	101.0	18.0	-1.0	-1.0	50
026f3f8b08	3.6130688247 37 E5E8	38.87970793	-77.11529015	42.0	101.0	23.0	-1.0	-1.0	50
026f3f8b07	3.6130688247 37 E5E8	38.8796842	-77.11539471	42.0	101.0	20.0	-1.0	-1.0	50
026f3f8b03	3.6130688247 37 E5E8	38.87969332	-77.11530435	42.0	99.0	20.0	-1.0	-1.0	50
026f3f8b0c	3.6130688247 37 E5E8	38.87969988	-77.11501041	42.0	96.0	16.0	-1.0	-1.0	50
0246c67e8	3.613068870175E8	0.0	0.0	-1.0	0.0	-1.0	-1.0	-1.0	0
0246c67e6	3.613070002456E8	0.0	0.0	-1.0	0.0	-1.0	-1.0	-1.0	0

10. Integrated Timeline

The following chronology consolidates the supplied examination findings and the case-report timeline. Where source documents conflict, the discrepancy is explicitly marked.

Date / Time	Source	Event	Forensic Significance
2012-06-06	Device log	iPhone activation / first use attributed to Tracy	Device profiling
2012-06-19	Email	Pat instructs Tracy to use alias email / secure communications	Planning / concealment
2012-07-03	SMS	Financial pressure concerning school tuition	Possible motive
2012-07-05	SMS	Carry and Tracy arrange meeting at Bubba's Grill	Flash-mob planning
2012-07-06	Email	Pat recruits King; Tracy's alias is copied	Stamp-theft planning
2012-07-06	SMS	Tracy and Carry coordinate meeting	Operational coordination
2012-07-08 12:41–13:01	Images / GPS	Stamp photographs captured at NGDC area	Reconnaissance / presence
2012-07-09	Email	Insurance documentation retained/sent	Insider knowledge / planning
2012-07-10	Email	King provides preparation list	Advance preparation
2012-07-10	SMS	Pat and Tracy discuss changing attachment format	Operational coordination
2012-07-11	SMS	Carry coordinates tablet transfer; Tracy says she will take it inside	Insider access
2012-07-11/12	SMS	Flash-mob/security coordination	Art-defacement planning
2012-07-15	Evidence	Forensic image identified / acquired in supplied materials	Evidence preservation

10.1 Timeline Interpretation

The timeline shows a progression from financial pressure and communications-security measures, through recruitment and reconnaissance, into operational coordination. The July 8 photographic activity is a central anchor because it provides both subject-matter evidence and location metadata that can be cross-correlated with the location database.

11. Evidence Correlation and Findings

A defensible forensic conclusion should be based on the convergence of independent artifacts. The following matrix summarizes the principal correlations established from the supplied materials.

Proposition	Supporting Artifacts	Independent Corroboration	Assessment
Device associated with Tracy	Device logs, owner name, accounts, phone number	Multiple device/account artifacts	High
Alias mailbox used for sensitive material	POP3 mailbox + messages + attachments	Account protocol and artifact paths	High
Stamp targets were photographed	Camera images + descriptions	EXIF timestamps/GPS	High
Device was at NGDC during photography	EXIF GPS	consolidated.db location records	High
Stamp theft was being planned	Recruitment email + preparation attachment	Tracy alias copied / related SMS	High
Flash-mob operation was being coordinated	SMS with Carry	Location/timeline correlation	Moderate-High
Financial pressure existed	SMS with Terry	Timing before planning events	High
Insurance fraud was the objective	Insurance docs + financial pressure	Requires external insurer/financial evidence	Moderate / hypothesis

11.1 Role Assessment

Person / Identifier	Assessment from supplied evidence	Confidence
Tracy Sumtwelve	Primary subject; device user; insider access; recipient/participant in planning artifacts	High
Pat Sumtwelve	Planning contact; recruited King; communicated with Tracy	High
Carry	Flash-mob organizer; coordinated meetings/equipment with Tracy	High
King	Recruited intended executor; supplied preparation list	High
Terry	Family member; financial-pressure context; no evidence in supplied material that she knowingly participated	Low involvement / likely unwitting

IMPORTANT DISTINCTION

The evidence supports attribution of communications and artifacts to accounts/devices. It does not, without additional corroboration, establish that every person associated with a message knowingly participated in a completed offense.

12. Conclusions

Based on the supplied forensic image, examination notes and case materials, the following conclusions are supported.

C1 – Integrity: The forensic image passed the documented MD5 and SHA-256 integrity checks.

C2 – Device attribution: The examined iPhone filesystem contains multiple identifiers and account artifacts consistent with Tracy Sumtwelve as the device subject.

C3 – Planning: Email and SMS artifacts provide substantial evidence of advance planning and coordination involving multiple participants.

C4 – Reconnaissance: The camera artifacts document photographs of relevant stamp material on July 8, 2012, with embedded GPS metadata.

C5 – Physical presence: EXIF GPS and iOS location artifacts independently corroborate the device's presence in the NGDC area during the photographic activity.

C6 – Insider access: The communications and work-account context support the assessment that the subject had access to information and facilities relevant to the investigation.

C7 – Financial motive: The SMS evidence supports financial pressure, while insurance documents demonstrate the high value of the relevant stamp lots.

C8 – Insurance-fraud theory: The combination is consistent with an insurance-fraud hypothesis, but the forensic image alone does not establish that an insurance claim was filed, paid, or intended as the definitive objective.

12.1 Evidence Quality

Evidence Type	Quality	Reason
Email	Excellent	Direct messages, sender/recipient metadata and attachment context
SMS	Excellent	Direct device-stored communications with timestamps/contacts
Photographs / EXIF	Excellent	Timestamped media with GPS and subject matter
Location data	Good	Strong corroboration; accuracy/method should be retained
Insurance documents	Excellent as artifacts	Directly recovered documents; authenticity should be externally verified

13. Recommendations

The following recommendations are framed as investigative next steps, not assumptions that the underlying allegations are proven.

Priority	Recommended Action	Purpose
High	Preserve and independently hash the original evidence image and working copies	Maintain integrity and reproducibility
High	Resolve serial-number, evidence-count and insurance-value discrepancies	Prevent avoidable evidentiary challenges
High	Validate insurance documents with the insurer / collection records	Establish authenticity and policy context
High	Review NGDC CCTV and access-control records for relevant dates	Independently corroborate physical presence and actions
High	Forensically examine relevant devices/accounts of other participants under proper authority	Corroborate communications and role attribution
Medium	Obtain provider-side records for the alias mailbox where legally authorized	Validate account access and chronology
Medium	Examine NGDC work systems available to the subject	Establish access to collection and insurance information
Medium	Review financial records under appropriate legal authority	Test the financial-motive hypothesis

13.1 Preservation Priorities

- Preserve the original forensic image in controlled evidence storage.
- Maintain a documented chain of custody for every evidence transfer.
- Preserve exported artifacts with original filenames, paths and cryptographic hashes.
- Preserve source screenshots/exports used to support key findings.
- Record software versions, analysis dates and examiner actions for reproducibility.

14. Chain of Custody and Evidence Handling

The supplied case report includes a chain-of-custody table, but the original seizure details are recorded as unknown. This report therefore reproduces the known handling milestones while clearly identifying the incomplete portion.

Date	Action	Personnel	Location / Note
Unknown	Device seizure	Law enforcement	Tracy's residence – source record does not provide a precise date
2026-05-30	Examination notes / assignment context supplied	Student examiner	Current report preparation context
[Source case date]	Forensic image received / imaged	As recorded in case material	Digitech Inc. forensic lab
[Source case date]	Analysis	As recorded in case material	Digitech Inc. forensic lab

CHAIN-OF-CUSTODY LIMITATION

The supplied case materials contain historical dates and a pre-existing analyst identity that do not match the current report-preparation context. This report does not fabricate missing custody events. Any real-world submission should replace bracketed fields with the actual evidence-transfer records.

14.1 Evidence Handling Statement

No claim is made in this report that the current examiner personally seized the device, created the original image, or witnessed the historical chain of custody. The integrity conclusion is limited to the documented hash match for the supplied image.

15. Evidence Register

ID	Artifact	Path / Identifier	Relevance
E-01	Forensic image	tracy-phone-2012-07-15-final.E01	Primary evidence source
E-02	Apple support log	/vol5/mobile/Library/Logs/AppleSupport/general.log	Device profiling
E-03	Alias mailbox	/vol5/mobile/Library/Mail/POP-coralbluetwo@hotmail.com@pop3.live.com/INBOX	Sensitive communications
E-04	SMS database	/vol5/mobile/Library/SMS/sms.db	Communications and contacts
E-05	Camera media	/vol5/mobile/Media/DCIM/100APPLE/	Photographic reconnaissance
E-06	Primary image	IMG_0050.JPG	EXIF/GPS evidence
E-07	Location database	/vol5/root/Library/Caches/locationd/consolidated.db	Movement/location corroboration
E-08	Insurance documents	documents.zip / three PDF artifacts	Stamp values and policy context
E-09	Preparation attachment	needs.txt	Advance planning context
E-10	Recruitment email	9F0508B8-04FB-490E-A7F0-3E23B0E7C59B.emlx	Planning / recruitment

15.1 Key Artifact Paths

```
/vol5/mobile/Library/Mail/
/vol5/mobile/Library/SMS/sms.db
/vol5/mobile/Media/DCIM/100APPLE/
/vol5/root/Library/Caches/locationd/consolidated.db
```

16. Stamp Insurance Documentation

Lot #	Stamp Description	Insurance Value	File Location
Stamp Insurance 1.pdf			
25	Armed Forces Reserve	\$43,000.00	Artifact: 9223372036854601274
26	Stamp of Kazakhstan2	\$29,000.00	documents.zip
27	BradyCo.	\$12,000.00	tracysumtwelve@gmail.com
Stamp Insurance 2.pdf			
11	Woman's Profile	\$31,000.00	Artifact: 9223372036854601272
12	Stamp of Kazakhstan	\$29,000.00	documents.zip
13	1929 Nepal	\$27,000.00	tracysumtwelve@gmail.com
Stamp Insurance 3.pdf			
1	Douglas MacArthur	\$35,000.00	Artifact: 9223372036854601271
2	Nederlands	\$30,000.00	documents.zip
3	Mongolia	\$24,000.00	tracysumtwelve@gmail.com
TOTAL ESTIMATED VALUE:		\$260,000.00	

Appendix A – Standards Mapping

The report structure below maps the major reporting controls to the guidance consulted for this exercise.

Reporting Control	Report Section	Guidance Basis
Evidence identification	3, 16	ISO/IEC 27037; NIST SP 800-86
Integrity / hashing	3	ISO/IEC 27037; SWGDE acquisition/examination guidance
Examination process	4	NIST SP 800-86; ISO/IEC 27042
Artifact provenance	5–9, 16	ISO/IEC 27042
Cross-source corroboration	10–11	ISO/IEC 27042
Limitations and discrepancies	13	ISO/IEC 27042
Chain of custody	15	ISO/IEC 27037; SWGDE

Reference Standards

- NIST SP 800-86, Guide to Integrating Forensic Techniques into Incident Response.
- ISO/IEC 27037:2012, Guidelines for identification, collection, acquisition and preservation of digital evidence.
- ISO/IEC 27042:2015, Guidelines for the analysis and interpretation of digital evidence.
- SWGDE, Best Practices for Computer Forensic Examinations.
- SWGDE, Best Practices for Computer Forensic Acquisitions.
- SWGDE, Best Practices for Digital Evidence Collection.
- SWGDE, Best Practices for Mobile Device Evidence Collection, Preservation, Handling, and Acquisition.

Appendix B – Analyst Declaration

I certify that this report has been prepared from the evidence and case materials supplied for the examination and that findings have been distinguished from interpretations where appropriate. I have not intentionally altered the supplied evidence image.

I further state that missing historical acquisition or chain-of-custody information has not been fabricated. Where source materials contain conflicting values, those conflicts are disclosed in the report.

This document is an examination report for the supplied case exercise. It is not a legal opinion and should not be treated as a substitute for jurisdiction-specific evidentiary review.

Field	Entry
Examiner	Ayham Belal Megdadi
Organization / Course	Digital Forensics - ANU
Report Date	30 May 2026